

Analysis of Agricultural Finance in India:
Systematic Resilience, Governance Challenges,
and Cross-Country Comparison

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I. Introduction

Agriculture plays a vital role in India's socio-economic structure, serving as both the primary source of livelihood for millions and the backbone of national food security. Its strategic significance is especially evident in its role as the largest employer in the country. As of 2021 – 2022, agriculture accounted for 45.5% of India's workforce, far exceeding any other sector¹. However, this overwhelming employment share contrasts with the sector's economic output which contributes only 16%–17% to the national GDP.

The imbalance between employment rate and GDP reflect structural inefficiency, low productivity, and minimal mechanization. Over the past decade, the number of agricultural workers had still risen by nearly a third since 2017, driven in part by limited employment opportunities and the economic aftershocks of events such as the COVID-19 pandemic². Despite these growing numbers, agricultural productivity per worker remains the lowest compared in manufacturing or services. For instance, while the gross value added per worker in manufacturing exceeds ₹3 lakh and service exceeds ₹4 lakh, agriculture struggles to cross ₹1 lakh that indicates workers are stuck in unproductive job environments and limited opportunities².

The sector's internal composition further highlights its complexity. While crop continues to dominate with over 54% share in gross output, allied activities such as livestock (31.2%), forestry (7.7%), and fisheries (7%) are not yet emerging as significant growth drivers³. The data show that Indians still rely on crops rather than any other types in agriculture. These allied areas also underdeveloped and often overlooked in policy and financial planning further exacerbate the disparities.

Along with all of these, agriculture is not merely a legacy sector but a critical domain requiring renewed policy attention, financial innovation, and inclusive development strategies. Agriculture makes up a large portion in people's lives and needs strong reforms especially on the gap between how many people it employs and how little it produces. Through effective agricultural finance planning, it can support disadvantaged communities like women or farmers and help the sector cope with changing economic and climate conditions.

II. Issues in Agricultural Finance in India

India's agricultural sector faces a complex mix of structural, social, and environmental challenges that make agricultural finance essential. Based on the agriculture census, India landholding in average drops more than 50% to 1.08 hectare per person (Fig. 1)⁴. This decrease in landholding per person alongside increase in the number of landholders show more and more

people working on the same land with low productivity or disguised unemployment. Workers are stuck from the dependence on agriculture, unable to move to a more productive sector. The deterioration of medium and large operations and areas also slows growth in income and widens the disparity between economically weaker and stronger states.

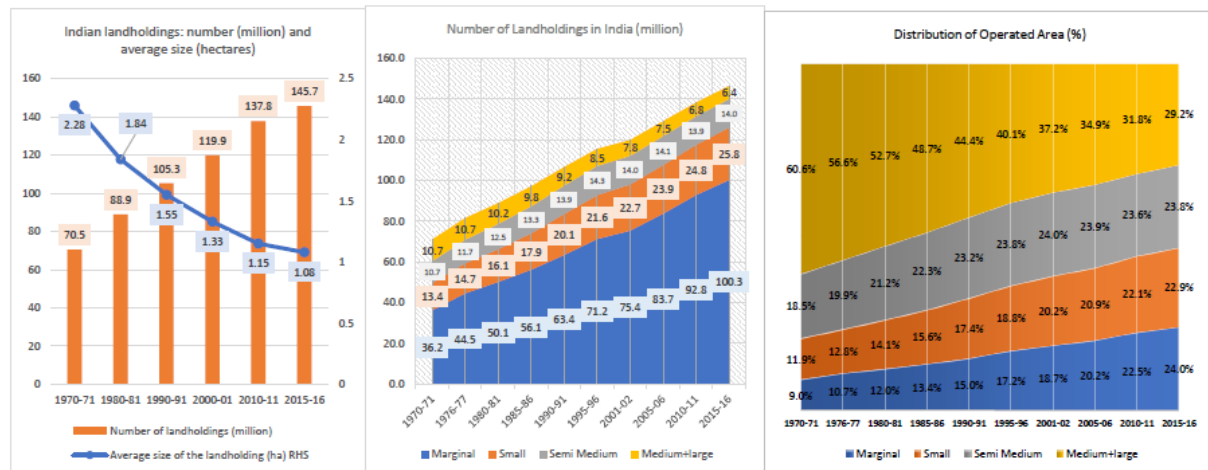


Fig. 1. Indian Agricultural Operational Holdings⁴

Additionally, the country's continued policy focus on rice and wheat, a legacy of the Green Revolution has led to an over-subsidization of cereals at the cost of diverse and more nutritious crops like pulses and millets⁵. The production of millet and minor cereals almost stationary with land cultivation are decreasing over the years. This trend in food availability influences the consumption behavior by people that change over time. Nowadays governments are still struggling to make the shift from rice to more diverse agricultural production prior to the green revolution²⁵.

India's structural issue of regional imbalance in technology and planting irrigation makes the country infrastructure distribute unequally. For example, the eastern states such as Bihar, Odisha, and Jharkhand struggle with poor irrigation and lag behind agriculturally while advanced northern states like Punjab and Haryana are the ones with advanced agriculture technologies⁷. Alongside with infrastructure imbalance, an urgent issue from over planting of rice from Green revolution haunts the water condition in the country that leads to scarcity in some key areas. With 54% of India's land facing high to extremely high water stress makes agriculture unsustainable⁶. This water stress also causes excessive groundwater depletion and puts the country's water supply at risk. The environmental pressures such as soil degradation, land desertification, and climate change further strain the system and increase uncertainty risk with each passing season.

On the social front, agriculture is undergoing a major demographic shift. As more men migrate to cities for better-paying jobs, women are stepping into primary agricultural roles

referred to as the feminization of agriculture. The share of women in agriculture by 2023 - 2024 was approximately 64.4 percent⁸. Even though they have the biggest share in the workforce, only 13 percent of the women really hold land titles in their name⁸. As a result, there's great disparity in wages between genders and women enjoy little decision in income. Most women's growing contributions as farmers also lack access to credit and training which limits their ability to improve autonomy.

In this context, agricultural finance becomes a lifeline. Timely and affordable credit helps farmers buy seeds, fertilizers, and equipment during planting seasons and also allows them to invest in long-term improvements such as irrigation, mechanization, and climate-resilient practices⁹. But access to formula finance remains limited especially for smallholders, tenant farmers, and women because these groups often lack collateral or stable income, hence, traditional lenders view them as high-risk borrowers. Thereby, the very people who need financial support are often excluded from the system.

To address this, India's financial policies must move beyond broad subsidies and introduce more targeted inclusive mechanisms. They could include interest subventions, flexible repayment terms, and credit guarantees that encourage banks to lend to high-risk groups. In short, without strong and inclusive financial systems, the agricultural sector cannot overcome its structural weaknesses, reduce rural poverty, or prepare for growing environmental risks.

III. Background of India's Agricultural Finance System

India's agricultural finance system is designed to support the country's vast rural economy. Anchored by government regulation and key financial institutions, it plays a central role in promoting rural development, reducing poverty, and boosting agricultural productivity. India's system functions through a multi-agency model involving apex institutions like the Reserve Bank of India (RBI), the National Bank for Agriculture and Rural Development (NABARD), and a network of three different types of banks (Fig. 2)¹⁰.

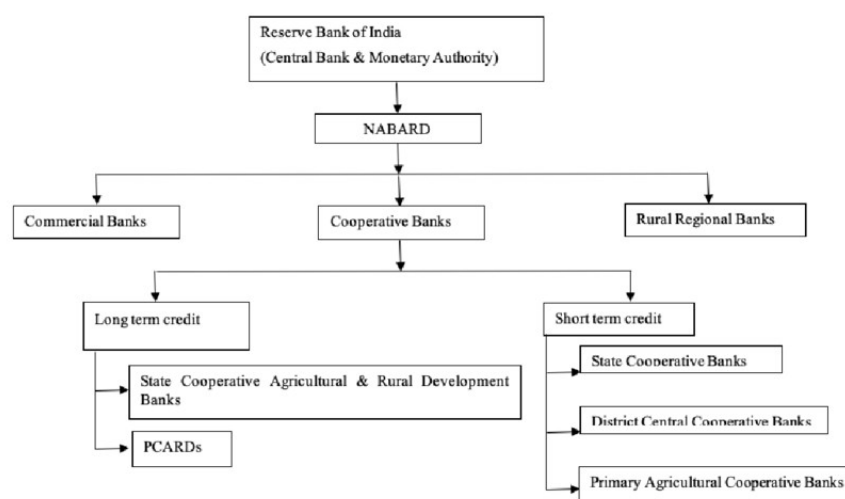


Fig. 2. Indian Agricultural Finance Structure¹¹

The RBI, India's central bank, is the chief regulator of the banking system. In agricultural finance, it performs three key roles. The first one sets lending mandates for banks to direct credit to agriculture under the Priority Sector Lending (PSL) framework. Next, RBI must monitor all rural and vulnerable populations to have access to formal finance. Last, regulating PSL Compliance to commercial banks which allocate at least 18 percent of their Adjusted Net Bank Credit (ANBC) to agriculture with a sub-target 8 percent for Small and Marginal Farmers (SMFs)¹⁰.

Below the RBI, NABARD became the apex development bank focused on rural credit which was established by an Act of Parliament in 1982. NABARD key responsibilities include refinancing loans and managing the Rural Infrastructure Development Fund (RIDF). NABARD refinancing operations will include Cooperative banks, Commercial banks, and Regional Rural Banks (RRBs) for lending to farmers from top to down in supporting long-term capital investment in projects like land improvement, irrigation, mechanization, and storage.

Beneath NABARD, agricultural credit is distributed via a multi-layered institutional network: Commercial Banks, Cooperative Banks, and Regional Rural Banks (RRBs). Commercial Banks include both public and private banks such as State Bank of India and Axis Bank. They are indeed crucial in providing large-scale loans in agribusiness and infrastructure and actively participating in PSL mandates. However, these banks often prefer lending to lower risk clients as they prioritize maximization of the profit they can get from the lending activity. Even so, under the PSL mandate, regulatory burden is imposed to push banks to meet lending targets even when risk is high in some of the line of credit¹².

Another institutional network is the Cooperative Banks that operate through a three-tier system of which they are the State Cooperative Banks (StCBs), District Central Cooperative Banks (DCCBs), and Primary Agricultural Credit Societies (PACS)¹³. Together,

they serve grassroots rural communities with focus on short-term and small-ticket loans for seeds, fertilizers, and daily needs. These types of banks prioritize community service to help all the member rather than generate profit. While essential for last-mile credit delivery, cooperative banks usually face problems of inefficiency, political interference, and bad loans. Finally, Regional Rural Banks (RRBs) were created to serve underbanked and remote rural areas. The RRBs are jointly owned by the Central Government (50%), State Governments (15%), and sponsoring banks (35%)¹⁴. They offer credit, savings, and insurance products in order to maximize productivity and blend local knowledge with formal banking infrastructure.

Moving on to the financial instruments that are used by the government, PSL is one of most regulatory tools to ensure that critical sectors, especially agriculture, receive adequate access to formal credit. Under PSL mandates, 18 percent of ANBC must be allocated to agriculture with 8 percent specifically reserved for Small and Marginal Farmers (SMFs). This includes loans for production, infrastructure, agri-startups, and post-harvest management. Loans also aim to promote value-chain finance such as funding for agro-processing and Farmer Producer Organizations (FPOs). However, this tool is not without setbacks. PSL can create tension within the financial system as commercial banks may end up focusing more on meeting volume targets than on careful risk evaluation. Banks rely heavily on Priority Sector Lending Certificates (PSLCs) to trade compliance rather than expanding genuine credit to the sector.

Introduced in 2016, PSLCs are a market-based mechanism that allows banks to trade their PSL achievements. In proactive, a bank that exceeds its priority sector targets can sell PSLCs to another bank that has fallen short. This system helps ensure overall compliance with PSL requirements while reducing pressure on individual banks to meet targets entirely through direct lending. At the same time, PSLCs create market signals by indicating which priority sectors are underserved and lacking credit. PSLCs are divided into four categories: General (41%) which covers broad sectors like housing, education, and MSMEs; SMFs (34%) aimed at improving credit access for farmers owning less than two hectares; Micro Enterprises (14%); and Agriculture (11%).

After setting all the lending mandates, the government also pushed forward the scheme for farmers to encourage borrowing. The Interest Subvention Scheme (ISS) complements PSL by lowering borrowing costs for farmers and encouraging timely loan repayment. Under this scheme, eligible borrowers receive an interest reduction of around 2-3 percent if they repay their loans on time. By directly rewarding good credit behaviour, the ISS helps reduce the effective cost of agricultural credit while also improving repayment discipline within the rural financial system.

Alongside these measures, the Indian government is setting up the Rural Infrastructure Development Fund (RIDF) which is managed by NABARD and plays a key role in financing rural infrastructure projects. The fund is primarily financed through contributions from commercial banks that fail to meet their Priority Sector Lending (PSL) targets. In this way, RIDF acts as a corrective mechanism within the financial system that ensures unmet priority sector obligations are redirected toward long-term rural development. By channeling capital into essential infrastructure, RIDF supports agricultural productivity and improves overall rural connectivity and market access.

On the other hand, one of the most important risk-management instruments in Indian Agriculture is the Pradhan Mantri Fasal Bima Yojana (PMFBY). This government-backed crop insurance scheme is designed to protect farmers from the financial impact of natural disasters and unpredictable crop losses. PMFBY is especially important for small and marginal farmers who are more vulnerable to climate-related shocks and lack savings to recover from failed harvests. Under the scheme, farmers pay a low fixed premium: 2% for Kharif crops, 1.5% for Rabi crops, and 5% for horticulture or commercial crops. The scheme also offers compensation for yield loss which consist of prevented sowing (25% of the sum insured) and post-harvest losses (within 14 days) and it is available to all farmers growing notified crops in participating states.

Another key policy instrument is the Kisan Credit Card (KCC) scheme which was introduced to make credit more accessible and flexible for India's farming community, especially small and marginal farmers. By providing short-term loans with affordable interest rates, KCC helps ensure that farmers have reliable financial support throughout the agricultural cycle from planting to harvesting. It works like a revolving credit line, allowing farmers to withdraw funds when needed without going through complex loan procedures each time. The scheme supports a wide range of short-term needs including the purchase of seeds and fertilizers, machinery, maintenance, and other operational expenses. Loans under KCC carry an interest rate of 7% which can be reduced to 4% with timely repayment²¹. The scheme targets not only crop farmers but also fishers and dairy or poultry farmers, helping ensure steady access to affordable credit across different agricultural activities.

Given the sector's low productivity, disguised unemployment, fragmented landholding, and environmental vulnerability, access to affordable and inclusive finance is vital to agriculture. Adequate agricultural finance supports investment in productivity-enhancing inputs and rural infrastructure to promote resilience through better risk management and empowers small and marginal farmers who form the backbone of Indian agriculture. It also

plays a growing role in supporting women in agriculture who increasingly shoulder both productive and household responsibilities within rural economics. Beyond the farm level, access to finance helps strengthen rural value chains, improve market linkages, and enhance national food security. Together, the institutional and policy ecosystem led by the RBI and NABARD supported by PSL mandates, cooperative networks of banks, and other instruments ensure that agricultural finance functions not merely as a source of credit but as a development lifeline for India's rural economy.

IV. Production and Financial Risk

India's agricultural sector faces not only structural financial challenges but it also has to deal with unpredictable shocks that threaten its stability and throw the entire system off. These shocks come in two broad forms: economic disruptions such as farm loan waivers and demonetization, and climate-related risks including floods, droughts, and extreme weather events. Together, each of these events exposes the vulnerabilities in the agricultural finance system and highlights the need for more resilient credit infrastructure.

1. Farm Loan Waivers

Farm loan waivers are a recurring political intervention aimed at relieving farmer debt during times of distress. India has seen major nationwide waivers in 1990 and 2008 (US \$16-17 billion dollars), not to mention there are more recent state level programs implemented by governments in individual states in India like Maharashtra, Uttar Pradesh, Punjab and others. These waivers often emerge around election cycles as political promises which then raises concerns about their use as political tools rather than financial tools.

While these policies provide immediate relief by reducing debt burdens and preventing farmers suicides or defaults, they introduce a moral hazard. Many farmers, after receiving a waiver, begin to expect future loans to be forgiven and reduce their incentive to repay future debts. Consequently, banks perceive small and marginal farmers as high-risk borrowers and become reluctant to lend to them leading to reduced access to formal credit. This cycle has long term consequences such as farmers increasingly depend on informal money lenders who typically charge really high interest rates hovering between 30 to 40% annually. Studies also show around 60% of farmers lack access to formal credit and financial exclusion worsens over time especially for the very farmers the waivers aim to help.

Ultimately, loan waivers distort the credit market. Instead of strengthening the agricultural finance ecosystem, they weaken the relationship between farmers and lenders that

undermine financial discipline and risk assessment. In fact, coming from an expert's perspective, economists warn that waivers only politically popular have actually derailed the economy. They argue that the same funds would be better used for long term investments in rural infrastructure and access to formal credit.

2. Demonetization (2016)

In November 2016, the Indian government invalidated all ₹500 and ₹1,000 currency notes, effectively removing 86% of cash from circulation¹⁶. While the move was intended to tackle corruption and black money, it had severe unintended consequences for the agricultural sector which operates largely on cash. India's rural economy where 90% of transactions are cash-based was hit particularly hard¹⁷. It started off with farmers who could not buy seeds, fertilizers, or pay laborers during peak Rabi planting season. Markets also respond with sudden drop in demand, sellers and farmers just trying to get some money are forced to sell their goods at a very low price leading to the country's price crashes. Tons of waste of fruits and vegetables all over the market due to their perishable characteristics that can't be stored for a longer time. Furthermore, this highlights that small farmers faced severe liquidity shortage and halting their next production cycle.

According to a study published in the American Journal of Agricultural Economics, agricultural trade at that time fell by over 16% in the months following demonetization especially for Kharif crops and areas far from formal banking systems¹⁵. Thus, to mitigate the damage, government introduced emergency relief where they extended loan repayment deadlines by 60 days, allowing farmers to still qualify for interest rebates, waived interest on short-term cooperative loans for two months (worth ₹660 crores), and allowed farmers to use old currency notes to buy seeds and fertilizers from government-approved sellers. Despite these efforts, demonetization exposed the fragility of India's rural financial systems and emphasized the need for greater financial inclusion, digitalization, and banking access in agriculture.

V. Literature Review

Academic literature on agricultural finance in India reveals long-standing systemic issues ranging from ineffective implementation and governance failures to the unintended consequences of politically motivated interventions¹⁸. One study also emphasizes the growth rate is highest during the period 1980–1989 (almost above 4%)²⁶. After 1990 the average annual growth rate is declining. Despite the number of initiatives aimed at improving rural credit

access, empirical studies consistently highlight gaps in institutional efficiency, sustainable outreach, and financial inclusion.

Historical analyses show a cyclical pattern in India's rural financial policymaking. Over time, India has introduced many schemes to support rural finance like microfinance programs, cooperative banks, and crop insurance. But researchers note a pattern, many of these ideas resurface under new labels and in new forms, without solving the deeper problems. The core issue is not a lack of good ideas or even lack of money, it is actually how these programs are implemented. Weak governance, low accountability, and poor follow-through mean that even the best-designed policies often fail to make a long-term difference²⁸. Until these structural issues are fixed, India's rural finance system will keep facing the same old challenges.

A significant portion of rural credit delivery in India depends on Cooperative Banks and Regional Rural Banks. However, academic evaluations expose deep flaws in their operational efficiency. Using Data Envelopment Analysis (DEA), researchers found a continuous decline in the technical efficiency of cooperative banks between 2013 and 2019, the reasons behind this are the rise of unpaid loans, high Non-Performing Assets (NPAs), poor loan recovery mechanisms, misallocation of funds, and even political interference¹³. On top of that, these banks face high costs and often don't have enough capital to fall back on when things go wrong.

Given that Cooperative Banks serve over 120 million rural clients, their inefficiencies pose a serious threat to financial inclusion in rural India²⁷. Studies emphasize that fulfilling lending targets under Priority Sector Lending (PSL) norms alone is not enough to overcome this problem. Sustainable improvement requires deeper reforms in bank governance, credit monitoring, and risk management frameworks. Similarly, Regional Rural Banks were created to reach underserved rural areas in India. Many of these banks are struggling with high operating costs and limited capital they have to lend to the farmers. The enforcement of lending targets makes institutions fragile towards bankruptcy. Concern around structural reform of the rural credit is needed by modernize the service and plan to sustainable improvement for long term financial viability.

VI. Cross Country Comparison: India and Brazil

A comparative look at agricultural finance systems in India and Brazil offers valuable insight into how institutional design and policy direction can shape rural credit outcomes. While both are large developing countries with significant agricultural sectors, their approaches

to agricultural finance differ notably, especially in terms of structure, delivery mechanisms, sustainability focus, and funding sources.

1. Institutional Roles and Structure

In Brazil, agricultural finance is largely driven by BNDES (Banco Nacional de Desenvolvimento Econômico e Social), which operates as a national development bank. It provides long term, investment-oriented loans for activities such as purchasing modern machinery, building irrigation systems, or adopting new technologies. BNDES plays a direct role in shaping Brazil's agricultural modernization by acting as a government-backed investor in rural development.

In contrast, India's apex institution for rural credit, NABARD (National Bank for Agriculture and Rural Development) plays more of a supervisory and refinancing role. Rather than lending directly to farmers, NABARD provides financial support and refinancing to Cooperative banks, Regional Rural Banks (RRBs), and other intermediaries that serve farmers on the ground. While it ensures the flow of rural credit, its influence is more indirect and focused on facilitation rather than investment.

In short, this comparison highlights a key institutional difference between the two countries: Brazil's BNDES is a direct, long-term investor in agricultural development, whereas India's NABARD acts as an indirect facilitator that supports rural credit through a layered banking network.

2. Credit Flow Mechanism

Brazil's rural credit system relies heavily on indirect lending mechanisms. About 99% of BNDES loans are disbursed through intermediary institutions, such as local banks and cooperatives which have no mandates to follow for lending activity to the farmers. These institutions carry the credit risk and tailor loans based on their knowledge of local farming conditions. This model decentralizes risk and helps maintain strong borrower-lender relationships.

India, on the other hand, uses a target-driven approach. Under the Priority Sector Lending (PSL) policy, the Reserve Bank of India mandates that commercial banks allocate 18 percent of their adjusted net bank credit to agriculture. This direct lending strategy ensures that rural sectors receive a minimum share of credit. However, it can sometimes lead to inefficiencies when credit is allocated to meet quotas rather than actual demand. The credit often flows more as a policy obligation than as a market-driven response to farmers' real needs.

3. Programs and Strategic Design

Brazil's agricultural credit system is programmatic and highly specialized. BNDES operates through a wide array of structured credit lines or programmes, each tailored to a specific goal. For instance, MODERFROTA supports the acquisition of farm machinery and tractors, while MODERINFRA invests in rural infrastructure like irrigation systems. INOVAGRO funds agricultural innovation and tech adoption and the ABC Program (Low Carbon Agriculture) promotes sustainable practices by offering incentives for eco-friendly farming methods. In addition, PRONAF targets family farming and smallholder.

Beyond these examples, each program is designed with clear sectoral goals and farmers must apply for each loan separately based on their specific needs. This targeted design promotes strategic investment and accountability.

On the contrary, India follows a broader scheme-based approach. Programs like the Priority Sector Lending (PSL) have no specific sectoral goal, the Kisan Credit Card (KCC) only provide short-term working capital. Other schemes like PMFBY for crop insurance and the Rural Infrastructure Development Fund (RIDF) aim to support risk management and rural infrastructure indirectly. These schemes are bundles in nature, lacking the granularity and sector-specific targeting that could be seen in Brazil.

4. Focus on Sustainability

One of the most striking differences between the 2 systems lies in their approach to sustainability. Brazil has integrated environmental goals directly into its credit policy, programs like BAC and INOVAGRO offer lower interest rates to farmers who adopt low emission technologies or implement soil recovery techniques²³. In this way, sustainability is not treated as an add on but as a core criterion for financial support. Farmers are incentivized to adopt eco-friendly practices through cheaper loans and environmental performance become part of the credit decision²².

In India, sustainability is still largely disconnected from credit policy. While programs like PMFBY and RIDF may support climate resilience indirectly, they are not structured to promote sustainable agriculture explicitly. Credit allocation still emphasizes outreach and access rather than environmental outcomes.

Thus, Brazil's rural finance is not just about reaching farmers, it's about influencing how they farm. India's system, though inclusive, is yet to incorporate sustainability into its financial logic.

5. Funding Sources and System orientation

Brazil's rural credit ecosystem is a public-private partnership model. About 65% of credit comes from private sources such as commercial banks, cooperatives, and equipment manufacturers, while 35% is funded publicly through BNDES and government programs¹⁹. This blend creates a more flexible and market-responsive system.

India's funding is primarily public or mandated. Most rural credit flows either through government schemes or as a result of regulatory obligations like PSL. While this ensures financial inclusion, it also makes the system more dependent on government budgets, political cycles, and short-term priorities.

Brazil and India also share contrast on their achievements. Brazil's strategy has yielded significant long-term gains. Since the mid 1990s, BNDES's credit to agriculture has increased more than 5 times. Key surges in lending correspond with major policy initiatives such as the expansion of farm mechanization programs in the early 2000s and the roll-out of sustainability-linked credit lines around 2013 to 2014. Even with economic slowdowns after 2015, credit volumes have remained historically high, showing Brazil's continued reliance on institutional finance for agricultural transformation.

India's rural finance, in contrast, remains more cyclical and reactive, often influenced by annual budgets, monsoon outcomes, or political pressures. Credit is typically short-term and consumption-oriented (seasonal loans or crop insurance payouts) rather than capital-intensive and investment-led.

VII. Cross Country Comparison: India and Taiwan

Comparing agricultural finance systems in Taiwan and India offers valuable insight into how institutional design and policy direction can shape rural credit outcomes. Both systems differ in many ways including their approaches in terms structure, delivery mechanisms, loan focus, and funding sources.

1. Institutional Roles and Structure

Taiwan's agrifinance system is centered around two main institutional levels, which were established under the "Agricultural Finance Act" in 2004²⁰. The purpose of the act is to provide sound management of agricultural finance institutes, ensure the rights of depositors, and promote the economic development of agriculture and fishery villages. The act refers to

two-tiered levels institutions: Credit Departments of farmer's and fishermen's associations (the main local-level financial units), and The Agricultural Bank of Taiwan.

Local-level Credit departments manage individual farmers' and fishermen's associations. Responsible for deposits and lending tied to many causes of agriculture, forestry, fisheries, and livestock. These units are required to keep their accounting process different from the association's other activities to ensure clearer governance and supervision. The Agricultural Bank of Taiwan acts as the system-level pillar, responsible for guiding, funding, monitoring, and stabilizing the network of Credit Departments. On top of both institutions, the authorities in-charge of the regulation: COA of the Executive Yuan or now Ministry of Agriculture along with the relevant local government authorities²⁴.

India, on other hand, is characterized by multi-tiered governance. Central authorities and oversight being the Reserve Bank of India (RBI), NABARD, and various banking entities. The system is also largely shaped by the Priority Sector Mandate, which required their commercial banks to allocate 18% of their Adjusted Net Bank Credit (ANBC) to the agriculture sector with a minimum share of 8% must be specifically reserved for small and marginal farmers. In contrast, Taiwan's approach relies less on lending agriculture credit but rather creates a different channel to support and finance the rural institutions.

2. Funding Sources

Taiwan's agriculture finance funding system mainly comes from a mixed set of funding sources, as established under the Agricultural Finance Act. At the bottom level, the Credit Departments of farmer's and fishermen's associations obtain their funding through deposit mobilization from local routine lending. Furthermore, Agricultural Bank of Taiwan (ABT) manages these funds by pooling rural savings at the system level and later reallocates the liquidity across the Credit Departments, providing funding as local units face a shortage and is involved in larger-scale agricultural funding projects that individual Credit Departments cannot handle by themselves.

Many policy-oriented funds in Taiwan are supported by the government through the Agricultural Development Fund. Each budget fund is decided through priority by the Ministry of Agriculture and executed by ABT. The funding source makes Taiwan more government-driven because public funds actively help finance and subsidize the agriculture programs. While India relies more on bank-driven regulatory mandates, each bank is required to provide funds and meet the credit quotas. Government support only comes through subsidies, interest support, or refinancing rather than directly fund most of the loans.

3. Delivery Mechanisms

In Taiwan, the delivery mechanism is separated through both local-level and system level. At local-level, Credit Departments of farmer's and fishermen's associations act as primary delivery channels by providing direct deposit and loan services to individual farmers, fishermen, and rural communities. The lending activities are later supported by ABT as they coordinate delivery to each credit department by providing liquidity support, supervising bigger credit cases, guiding operational standards, and disbursed policy-oriented agricultural finance flows from the Ministry of Agriculture.

By contrast, India has a more diversified channel. Through their commercial banks, regional rural banks (RRBs), and cooperative banks, in which, all entities are required to serve and deliver the agricultural credit, so the delivery is more well-spread across several types of institutions.

4. Loan Focus

Taiwan's agriculture system covers modernization-focus loans by supporting technology adoption and new production models. Financing is used for smart agriculture investment such as greenhouse control systems, equipment upgrading, and digital farm management tools. Credit also supports production transformation by helping farmers to shift through more specialized production models with all oriented to sustainability. These priorities are also reinforced by policy-oriented loans programs delivered by the Ministry of Agriculture.

India, on other hand, focused on broad and multi-purpose loans, covering both short-term production needs and wider rural investment. The agriculture credit is not focused on one purpose or another, the banks are free to lend to the farmers, whether it is for farm machinery, irrigation, rural infrastructure, and agriculture-linked MSMEs. Other examples include crop loans (KCC).

VIII. Conclusion

India's agricultural finance plays an important role in the economic, social, and environmental sector especially in shaping the nation's rural livelihoods, food security, and economic development. Although almost half of India's workforce is in the agriculture sector, the productivity of agricultural workers still remains lower than in manufacturing or services which make agriculture contribute a disproportionately small share to GDP. This indicates the imbalance of structure, the problems of inefficiencies, and underdevelopment which tend to be

overlooked in policy and financial planning. This problem was compounded by climate vulnerability, regional inequality, and demographic shifts such as the feminization of agriculture that make access to inclusive and affordable finance indispensable for the sector's long-term viability.

India's Agricultural Finance System is anchored by the Reserve Bank of India (RBI) and NABARD and delivered through commercial banks, cooperative banks, and Regional Rural Banks. India credit system are mandates to expand credit outreach through several instruments such as Priority Sector Lending, Interest Subvention Scheme, Rural Infrastructure Development Fund, and Kisan Credit Card (KCC). These mechanisms have played an important role in promoting productivity-enhancing inputs and rural infrastructure, supporting many marginal farmers who form the backbone of Indian Agriculture, and strengthened national food security. However, the same problems keep persisting as climate and economic disruptions worsen many of the industry's vulnerabilities, reflected in the case of farm loan waivers and demonetizations.

The cross-country comparisons analysis with Brazil and Taiwan further highlight the limitations of the Indian Agricultural Finance system. Brazil's model demonstrates the investment-oriented loans, indirect lending mechanisms, and sustainable-focused credit. While Taiwan's institutionally centered system, modernization-focused lending have demonstrated how credit can be used not only to teach farmers but also to shape production choices, promote innovation, and include environmental outcomes into the financial-decision making.

Overall, an improved agricultural finance system must be created in India. The agricultural finance system must change from being reactive to problems to being a stable system that facilitates long-term agricultural investment while considering environmental issues. Agricultural finance must be more efficient and therefore must allow access to financing by farmers (particularly female farmers). Agricultural financing must link access to financial resources with best production practices and with economic and environmental vulnerabilities associated with extreme weather and other climate change issues. Furthermore, the government must decrease investments in short-term solutions and increase investments in long-term rural infrastructure development and innovation and ideas. Absent the necessary reforms to the agricultural finance system, agricultural financing in India will only serve as a temporary lifeline to rural communities; it can never be an engine for the transformation of India's rural economy.

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X. Work Division Sheet

Contributor	Roles and Responsibilities
張柔晶 (B12607032)	Information Collection; Data Collection; Writing First Draft (Sections I & II). Conducted the primary research on India's workforce and GDP distribution and drafted the foundational context for the Indian agricultural finance system.
莊慧期 (B12603033)	Data Collection; Information Collection; Writing First Draft (Section IV). Focused on the analysis of the disaster and synthesized the case studies.
李冰冰 (B12607027)	Editing and Revising; Information Collection; Writing First Draft (Section V). Led the academic literature review and ensured the integration of technical concepts into the case of India
曾福輝 (B12607028)	Organizing Final Report; Information Collection; Writing First Draft (Section VI). Responsible for the cross-country comparative analysis and the synthesis of final policy recommendations.
All Members	Coordinating Group Work; Revising the Report; Making Final Presentation. All members participated in the final review and internal peer-editing process to ensure narrative fluidity and technical accuracy.